WORKED SOLUTION

STEP I 1992, Question 6

Pure • Small Angle Approximations

STEP I 1992, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q6

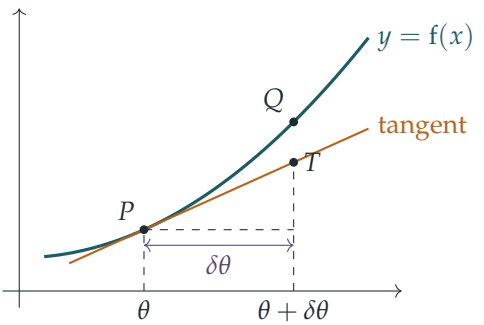
The key idea

The angle at the far point C is exactly ϕ, because $\theta + \phi$ is an exterior angle of triangle ABC. The sine rule then gives AC exactly. Since C is very far away, ϕ must be tiny, and the small-angle approximations (which are the first part's linear approximation in disguise) turn the exact formula into something simple. The error question is that same linear approximation used once more, this time with ϕ as the variable.

Solution

The linear approximation

Near the point $P = (\theta, \mathrm{f}(\theta))$ the graph of f is very close to its tangent line, which has gradient $\mathrm{f}'(\theta)$.



Move a distance $\delta\theta$ to the right. Along the tangent the height rises by (gradient)$\times$(run) $= \mathrm{f}'(\theta)\,\delta\theta$, reaching T. Along the curve it rises to Q, at height $\mathrm{f}(\theta + \delta\theta)$. The gap QT shrinks faster than $\delta\theta$ itself as $\delta\theta \to 0$, because

$$\frac{\mathrm{f}(\theta + \delta\theta) - \mathrm{f}(\theta)}{\delta\theta} \to \mathrm{f}'(\theta)$$

is precisely the definition of the derivative. So for small $\delta\theta$,

$$f(\theta + \delta\theta) \approx f(\theta) + f'(\theta)\,\delta\theta.$$

The distance of C

In triangle ABC the angle at A is θ, and the exterior angle at B is $\theta + \phi$. An exterior angle equals the sum of the two opposite interior angles, so

$$\angle ACB = \phi.$$

The interior angle at B is $\pi - (\theta + \phi)$, and $\sin(\pi - x) = \sin x$, so the sine rule gives the exact results

$$AC = \frac{a\sin(\theta+\phi)}{\sin\phi}, \qquad h = AC\sin\theta = \frac{a\sin\theta\,\sin(\theta+\phi)}{\sin\phi}.$$

Ask yourself Where does "h is very large compared with a" actually enter?

It tells us that ϕ is small. Rearranging the exact formula, $\sin\phi = a\sin\theta\sin(\theta+\phi)/h \le a/h$, so $h \gg a$ forces $\sin\phi$, and hence ϕ, to be tiny. (Seen from a very distant C, the baseline AB subtends almost no angle.) Now use the first part with $f = \sin$:

$$\sin(\theta+\phi) \approx \sin\theta + \phi\cos\theta \approx \sin\theta, \qquad \sin\phi \approx \sin 0 + \phi\cos 0 = \phi.$$

Therefore

$$h \approx \frac{a\sin^2\theta}{\phi},$$

as required, and

$$AC = \frac{h}{\sin\theta} \approx \frac{a\sin\theta}{\phi}.$$

(Keeping the $\phi\cos\theta$ term gives $AC \approx a\sin\theta/\phi + a\cos\theta$. The extra piece is only of the size of a, which is negligible beside a distance of order h.)

The effect of an error in ϕ

Treat AC as the function $g(\phi) = a\sin\theta\,\phi^{-1}$, with θ fixed because it is measured correctly. By the first part, an error $\delta\phi$ changes the estimate by

$$\delta(AC) \approx g'(\phi)\,\delta\phi = -\frac{a\sin\theta}{\phi^2}\,\delta\phi.$$

So the error in AC has size about $\dfrac{a\sin\theta}{\phi^2}\,|\delta\phi|$; overestimating ϕ underestimates the distance.

Ask yourself If we double a, the formula has a on top. Doesn't the error double?

No, because ϕ is not fixed: it depends on the baseline. The point C (and so h) stays the same, and from $h \approx a\sin^2\theta/\phi$ we get $\phi \approx a\sin^2\theta/h$, so doubling a roughly doubles ϕ. (θ changes only very slightly, since C is so far away.) Writing the error in terms of the fixed quantity h:

$$\frac{a\sin\theta}{\phi^2}\,|\delta\phi| \approx a\sin\theta\cdot\frac{h^2}{a^2\sin^4\theta}\,|\delta\phi| = \frac{h^2}{a\sin^3\theta}\,|\delta\phi|.$$

For the same instrumental error $\delta\phi$, the error in AC is inversely proportional to a. **Doubling the distance between the telescopes roughly halves the error.**

Checks

- Take $a = 1$, $\theta = \pi/6$, $\phi = 10^{-4}$. The exact formulae give $h \approx 2500.43$ and $AC \approx 5000.87$; the approximations give 2500 and 5000. The difference in AC is about $0.87 \approx a\cos\theta$, just as the extra term predicts.
- Computing C directly as the intersection of the two sight lines, for thousands of random a, θ and small ϕ, matches both approximations to within 0.1%, and the change in AC caused by a small $\delta\phi$ matches $-a\sin\theta\,\delta\phi/\phi^2$.
- For a fixed distant C, recomputing ϕ with baselines a and $2a$ and perturbing each by the same $\delta\phi$ gives error ratios of 2.
- Dimensions: $a\sin^2\theta/\phi$ is a length, since angles in radians are dimensionless.

Common slips

- Using $\sin\phi \approx \phi$ with ϕ in degrees. The approximation, like f′ for sin, needs radians.
- Concluding that the error doubles with a by holding ϕ fixed. The target fixes h, not ϕ.
- Missing that $\angle ACB = \phi$ and setting up coordinates instead. That works, but it takes longer.

Beyond STEP

This is triangulation by parallax, the method astronomers use to measure distances to nearby stars, with the Earth's orbit as the baseline. The error analysis is a first taste of error propagation: to first order, an error δx in a measurement becomes an error $g'(x)\,\delta x$ in anything computed from it. That idea, made precise by Taylor's theorem with a remainder term, is a central part of first-year analysis; see Abbott, *Understanding Analysis*.

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